

LM07 Company Analysis: Forecasting

1. Introduction.....	2
2. Forecast Objects, Principles, and Approaches.....	2
3. Forecasting Revenues	4
4. Forecasting Operating Expenses and Working Capital.....	6
5. Forecasting Capital Investments and Capital Structure	8
6. Scenario Analysis.....	9
Summary.....	12

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1. Introduction

This learning module covers:

- What to forecast, approaches to forecasting, and selecting a forecast horizon.
- Forecasting particular items: revenues; operating expenses and working capital; capital investments and capital structure.
- Use of scenario analysis in considering multiple outcomes.

2. Forecast Objects, Principles, and Approaches

What to Forecast?

Analysts can focus on different forecast objects such as:

- Drivers of financial statement lines: This concept was covered in the previous learning module. The net sales for Warehouse Club Inc were analyzed using two bottom-up drivers: the average number of stores opened and the average net sales per store. Net sales can be forecasted by forecasting these drivers individually and multiplying them. The advantage of this approach is that it improves the explanatory value of the forecast and may also improve accuracy.
- Individual financial statement lines: Instead of forecasting drivers, we can directly forecast individual financial statement lines. This approach is often used for lines without clear drivers, for less-material items, and for items that the analyst does not have a perspective on. For example: amortization expense, analysts may use the estimate provided by the management or simply assume that the quantity will remain the same in future periods.
- Summary measures: This includes items like free cash flow, earning per share, and total assets. The advantage of using these as forecast objects is efficiency. The disadvantage is reduced transparency, which makes it difficult to audit the forecast. This method is best suited when the summary measure is stable and predictable, or when issuer disclosures are severely restricted.
- Ad hoc objects: This includes items that may not have been reported on previous financial statements. Examples include the outcomes of a significant legal proceeding, a government regulatory action, a tax dispute, or a natural disaster.

An analyst's choice of forecast object depends on available information, efficiency, accuracy, explanatory value, and verifiability.

Focus on Objects That Are Regularly Disclosed

CFAI recommends using forecast objects that are either regularly disclosed or can be directly calculated using what is regularly disclosed.

Non-regularly disclosed information can be used to supplement forecasts but is problematic for direct use because forecasts cannot be confirmed in a timely manner.

CFAI also cautions against overly complex models, because they require more forecasts and take more time to update, often with no improvement in accuracy.

Forecast Approaches

For any object there are four general forecast approaches:

- Historical results (assume past is precedent): This approach uses past observed or calculated values as a forecast. This is an easy to implement approach and may be appropriate for companies operating in industries where the industry structure is not expected to change. The approach can also be used to forecast objects that are not material or that the analyst does not hold an opinion on. The approach is less appropriate for companies in cyclical industries because the future period could be at a different point in the business cycle.
- Historical Base Rates and Convergence: This approach uses an industry or peer group average or median, calculated over a long time period as a “base rate” for forecasting where an object will converge to over time. This approach is more complex than the previous approach because analyst discretion is required to select the object, the sample to calculate the base rate, and a time frame for convergence to the base rate. The approach may be appropriate for companies operating in well established industries such as banks, airlines, retailers etc. It is less appropriate for companies in changing or new industries.
- Management guidance: A company's management may publicly announce earnings, revenue, and other targets for the next quarter, year, or longer (known as management guidance or simply guidance). This guidance is often provided as a range (e.g. sales growth of 3%-5%) and is based on many sub-forecasts and assumptions by the management about macroeconomic growth, cost inflation, market share changes, exchange rates etc. Using guidance for forecasts may be appropriate when management has demonstrated a track record of reliable estimates.
- Analyst's discretionary forecasts: All other forecast approaches can be grouped under this category. This can include approaches based on surveys, quantitative models, probability distributions, and other unobservable inputs. Discretionary forecasts approaches are commonly used for companies in cyclical industries, companies that have no comparables, or companies undergoing a fundamental change like a shift in the competitive or regulatory environment.

An analyst's choice of forecast approach depends on the company's industry structure, sensitivity to the business cycle, and business model, as well as the reliability and availability of information.

Selecting a Forecast Horizon

The forecast time horizon is determined by the investment strategy for which the security is being considered, the cyclical nature of the industry, company-specific factors, and the preferences of the analyst's employer.

Long-term fund managers may focus their forecasting primarily on the next three to five years, whereas shorter-term managers may focus more on the next one or two quarters.

3. Forecasting Revenues

Forecast Objects for Revenues

Revenues can be forecasted using a top-down or bottom-up approach. Common top-down forecast objects include “growth relative to GDP growth” and “market growth and market share”.

Growth relative to GDP growth approach: In this approach, we first forecast the growth rate of nominal GDP. We then forecast the company's revenue growth relative to GDP growth. For example, we can assume that the company's revenue will grow at a rate of 150 bps above the nominal GDP growth rate.

The forecast may also be in relative terms. For example, if we forecast that the GDP will grow at 4%, and we believe that the company's revenue will grow at a 20% faster rate, then the forecasted increase in the company's revenue is $4\% \times (1 + 0.20) = 4.8\%$.

Market growth and market share: In this approach, we combine forecasts of growth in particular markets with forecasts of a company's market share. For example, assume Tesla is expected to maintain a market share of 1% in the automobile market. If the automobile market is expected to grow to \$30 billion in annual revenue, then Tesla's annual revenue is forecasted to grow to $1\% \times \$30 \text{ billion} = \300 million .

Examples of bottom-up drivers for revenue forecast include:

- *Volumes and average selling price:* Volumes and prices of the company's products are forecasted separately and multiplied to get a revenue forecast. For example: The revenues for a mutual fund can be forecasted based on the AUM and management fee rates.
- *Product-line or segment revenues:* Forecasts for individual products, product or business lines, geographic areas, or reporting segments are made and then aggregated into a total revenue forecast.
- *Capacity-based measure:* For example, a retailer's revenue may be forecasted based on same-store sales growth and sales related to new stores.
- *Return or yield based measure:* Forecasts based on balance sheet accounts. For example, a bank's interest revenue can be calculated as loans multiplied by the

average interest rate.

Separating Recurring and Non-Recurring Revenue or Revenue Growth

While making forecasts, non-recurring items and effects should be excluded. There are two types of non-recurring items and effects: those that are disclosed by a company's management and those that are not, which require analyst judgment to identify.

Examples of the first type include: the effect of changes in exchange rates, acquisitions/divestitures etc.

An example of the second type is the rapid increase in e-commerce sales during the COVID-19 pandemic. After things normalized, e-commerce sales as a percentage of retail sales began to decline.

Forecast Approaches for Revenues

Instructor's Note: The curriculum illustrates the four approaches to forecasting revenues through the 'Warehouse Club Inc' example. This information is presented as-is from the curriculum.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Top-down drivers (e.g., market size, market share)	Historical results	US retail sales excluding autos has grown on average by 3.4% each year for the last 10 years and the company's share of that (70 basis points in 2X19) has increased by 2 basis points per year. Nguyen can forecast revenues by assuming 3.4% market growth and 2 basis points of market share gain each year going forward.
	Historical base rates and convergence	Rather than assuming continued share gains, Nguyen may forecast that Warehouse's sales growth rate will converge to the market growth rate over the next five years, and forecast revenues by assuming a smooth decline in annual market share gains to 0 basis points by 2X25.
	Management guidance	If Warehouse provides annual or longer-term expectations for market size, market share, or sales, Nguyen can use these in her revenue forecast.
	Analyst's discretionary forecast	Warehouse has operated stores in some regional geographies for a long time. Nguyen may commission a study or survey to determine Warehouse's market share in its most established markets (which could be multiples of its all-in market share) and develop market size and share estimates by established/growing/new regions for Warehouse and aggregate them into a sales forecast.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Bottom-up drivers (e.g., stores, sales per store, membership, membership annual price)	Historical results	Warehouse has opened 4–6 new stores, grown net sales per store by 3.8%, and added ~450,000 members per year for the last 10 years. Nguyen could assume these same quantities going forward as the revenue forecast.
	Historical base rates and convergence	There are several mature retailers in the USA that ceased opening new stores several years ago. Nguyen may slow the rate of new-store openings gradually to zero in the USA.
	Management guidance	Management has provided guidance on new-store openings per year and the annual price of membership, which are bottom-up drivers for the revenue forecast.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for stores and members by determining stores per capita and the ratio of members to stores in Warehouse's more mature regional geographies and then applying these quantities, or fractions of them, to different regions where Warehouse is opening stores today. This approach would require a geographic analysis of stores and commissioning surveys to determine regional membership counts or the use of third-party or government data, if available.

Analysts must explicitly or implicitly incorporate their view on key risk factors in their revenue forecasts. Four key risk factors to consider for all companies are competition, changes in business cycle, inflation and deflation, and technological developments.

4. Forecasting Operating Expenses and Working Capital

Operating costs include cost of goods sold (COGS) and selling, general, and administrative expenses (SG&A). Disclosures about operating costs are often less detailed than revenue. Therefore, analysts are often forced to use more aggregated forecasted objects on the company level (rather than forecasting costs separately for different geographic regions, or business segments) or summary measures like EBITDA margins. However, forecasts for operating expenses should still be coherent with revenue forecasts. The choice of forecast object can vary depending on the forecast horizon.

Cost of Sales and Gross Margins

Since COGS are directly related to sales, we can forecast future COGS as a percentage of future sales. Analysts should understand the historical relationship between COGS and sales, and determine if this relationship is likely to decrease, increase, or remain unchanged.

$\text{Sales} - \text{COGS} = \text{gross margin}$. Therefore, COGS and gross margin are inversely related.

Some factors that analysts should consider while forecasting COGS are:

- Forecasting accuracy can be improved by forecasting COGS for the company's various segments or product categories separately.
- Determine if a company has employed hedging strategies to protect its gross margins. When input prices increase, COGS increase and gross margin decreases. However,

hedging strategies can help mitigate this impact.

- Examine gross profit margins of competitors. This can be a useful cross check for estimating a realistic gross margin. Any difference in gross margins for companies in the same segment must relate to differences in their business operations.

SG&A Expenses

As compared to COGS, SG&A expenses are less directly related to revenue.

SG&A can be broken down into two components – fixed and variable. The fixed components such as R&D expenses, management salaries, head office expenses, supporting IT and administrative operations tend to increase and decrease gradually over time. They do not fluctuate with sales.

On the other hand, the variable components such as selling and distribution costs are more directly related to sales. For example, when sales increase, the company may pay out higher bonuses to its salesforce, it may also hire additional salespersons.

Working Capital Forecasts

Working capital forecasts are typically made using efficiency ratios as the forecast objects, which are combined with sales and cost forecasts to project accounts receivables, inventories, accounts payable, and other current asset and liabilities.

Exhibit 9 from the curriculum illustrates the forecast of operating costs and working capital for Warehouse Club Inc.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross margin (on net sales)	Historical results	From 2X10 to 2X19, Warehouse's gross margin increased by an average of 2 bps per year. Such an increase may reflect increasing bargaining power over suppliers with scale, and Nguyen can use the same 2 bps per year increase in gross margin as a forecast.
	Historical base rates and convergence	Both the industry average gross margin and that of larger, more mature retailers such as Walmart are much greater than Warehouse's gross margin. But Warehouse's business model is to use a low gross margin on net sales to drive high-margin membership revenues. Therefore, Nguyen probably will not use this approach.
	Management guidance	If Warehouse management provides guidance on margins, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may formulate a discretionary forecast for gross margin by exploring the relationship between it, inflation/deflation in Warehouse's product categories, and management's desire for member savings vs. competitors to equal 5× the annual price of membership.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Selling, general, and administrative expenses (SG&A) as a percentage of sales	Historical results	SG&A, which captures the cost of operating stores and some merchandise distribution, is primarily composed of employee payroll. Relative to sales, it has remained mostly stable at or around 9%. Nguyen may forecast it to remain the same.
	Historical base rates and convergence	Warehouse's SG&A as a percentage of sales is already lower than that of larger retailer competitors (and substantially lower than the retail industry average), owing to its business model differences that emphasize store productivity and cost leadership. Nguyen will therefore probably not use this approach to forecast SG&A.
	Management guidance	If Warehouse management provides guidance on SG&A, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Since SG&A is primarily composed of employee payroll, Nguyen may forecast it based on projected employee productivity (e.g., employees to stores) and projected wage rates, which may be based on macroeconomic variables like projected economy-wide wage growth.
Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Working capital ratios (e.g., DSO, DOH, DPO)	Historical results	As of 2X19, Warehouse's DSO, DOH, and DPO were 3, 30, and 28, respectively. DSO and DPO have been stable for the last 10 years, while DOH has decreased by 7 days since 2X10. Nguyen might forecast a stable DSO and DPO and a modestly declining DOH over the next several years.
	Historical base rates and convergence	Warehouse's working capital ratios and cash conversion cycle are materially better than the industry average and some larger peers and have been for many years, owing to its business model differences. Nguyen will therefore probably not use this forecast approach, unless Warehouse changes its business model (e.g., it launches an e-commerce business).
	Management guidance	If Warehouse management provides guidance on working capital (e.g., a projection for the net working capital to sales ratio), Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for working capital by, for example, estimating net working capital to sales ratios by product category (e.g., fresh foods, packaged foods, non-food merchandise) and applying them to a net sales forecast also broken down by category.

5. Forecasting Capital Investments and Capital Structure

Forecasts for capital expenditure may differentiate between maintenance and growth capital expenditures. Forecasts for maintenance capital expenditures are usually based on depreciation and amortization expenses. Forecasts for growth capital expenditures are linked to a company's strategy, expansion plans, and revenue growth.

Forecasts for a company's capital structure take into account the historical leverage ratios and capital structure, the company's financial strategy, and capital expenditure forecasts.

Exhibit 12 from the curriculum illustrate these approaches for Warehouse Club Inc.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Capital expenditures as a percentage of net sales	Historical results	Warehouse has spent 2% of net sales on capital expenditures for the last 10 years, which Nguyen can use as the forecast going forward.
	Historical base rates and convergence	More mature retailers (e.g., Walmart in the USA) have all but stopped opening new stores, which has reduced their capital requirements to maintenance and new initiatives like e-commerce. If Warehouse does not intend to go to new geographies or launch an e-commerce business, Nguyen might forecast capital expenditures trending down to maintenance level, along with a decline in new-store openings per year, as Warehouse matures.
	Management guidance	Warehouse management provides guidance on new-store openings, which Nguyen can use to forecast capital expenditures (along with a maintenance capital expenditure forecast and a forecast for growth capital expenditures per new store).
	Analyst's discretionary forecast	Nguyen might instead use a discretionary forecast for capital expenditures by projecting a maintenance CAPEX amount (not disclosed by management) and growth CAPEX associated with new-store openings.
Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross debt to EBITDA	Historical results	For the last five years, Warehouse's ratio of debt to trailing twelve months EBITDA has been 1.10, which Nguyen can project going forward. The product of this ratio and projected EBITDA (from earlier forecasts of sales and operating expenses) is Warehouse's gross debt.
	Historical base rates and convergence	Warehouse's more mature competitors and peers use a greater amount of debt in their capital structure, and return more capital to shareholders with buybacks and dividends. Nguyen might forecast the same for Warehouse over time, with an increasing ratio of debt to EBITDA and percentage of free cash flow returned to shareholders.
	Management guidance	Warehouse's management may provide a target capital structure through a target debt to EBITDA ratio or credit rating, which Nguyen can use as the forecast.
	Analyst's discretionary forecast	Nguyen might instead make a discretionary forecast linked to her free cash flow forecast, projecting a debt to EBITDA ratio and return of capital to shareholders. Depending on Fyleton Investments' strategy, Nguyen might engage with Warehouse's management to advocate for such a capital structure.

6. Scenario Analysis

The final step in forecasting involves considering the possibility of various outcomes based on key risk factors and assessing their likelihood of occurrence.

Rather than develop single point estimate forecasts, analysts make several forecast scenarios that vary based on different outcomes with respect to key risk factors. To make investment

decisions, investors compare these scenarios to other analysts' forecasts for a company, as well as forecasts implied by current valuations.

Technological developments can change the economics of individual companies and entire industries. Sometimes a technological development results in a new product that threatens to cannibalize demand for an existing product. For example, introduction of tablets affected the demand for PCs. To estimate the impact on future demand for the existing product, analysts forecast the unit sales of the new product and multiply it by an expected cannibalization factor. The cannibalization factor can be different for different market segments. For example, the cannibalization factor of tablets is higher in the consumer segment than the non-consumer segment.

Technological developments can affect the demand for a product, the quantity supplied, or both. If technological development leads to lower manufacturing costs, then the supply curve shifts to the right as more quantity is produced at the same price. But if technology results in better substitute products, then the demand curve will shift to the left.

Example:

(This is based on Example 4 from the curriculum)

The pre-cannibalization PC shipment projections (in thousands) for next year are:

Consumer PC shipments	170,022
Non-consumer PC shipments	180,881
Total PC shipments	350,903

The forecasted tablet shipments (in thousands) for next year are:

Consumer tablet shipments	36,785
Non-consumer tablet shipments	1,686
Total tablet shipments	38,471

Assuming a cannibalization factor of 30% for the consumer segment and 10% for the non-consumer segment, calculate the post-cannibalization PC projections.

Solution:

Cannibalization in the consumer segment = $0.3 \times 36,785 = 11,036$

Cannibalization in the non-consumer segment = $0.1 \times 1,686 = 169$

Therefore, the post-cannibalization PC projections are:

Consumer PC shipments	$170,022 - 11,036 = 158,986$
Non-consumer PC shipments	$180,881 - 169 = 180,712$
Total PC shipments	339,698

To evaluate the impact of technological developments on operating costs and margins, we

first need to understand the breakdown between fixed and variable costs. If sales decrease because of cannibalization, then variable costs also decrease. However, the fixed costs remain unchanged.

Since very few companies provide a breakdown of fixed versus variable costs, we can estimate them using the following formulas:

$$\begin{aligned}\% \text{ Variable Cost} &\approx \frac{\% \Delta (\text{cost of revenue} + \text{operating expense})}{\% \Delta \text{revenue}} \\ \% \text{ Fixed Cost} &= 1 - \% \text{ Variable cost}\end{aligned}$$

Summary

LO: Explain principles and approaches to forecasting a company's financial results and position.

Analysts can focus on different forecast objects such as:

- Drivers of financial statement lines
- Individual financial statement lines
- Summary measures
- Ad hoc objects

An analyst's choice of forecast object depends on available information, efficiency, accuracy, explanatory value, and verifiability.

LO: Explain approaches to forecasting a company's revenues.

For any forecast object there are four general forecast approaches:

- Historical results
- Historical base rates and convergence
- Management guidance
- Analyst discretion

An analyst's choice of forecast approach depends on the company's industry structure, sensitivity to the business cycle, and business model, as well as the reliability and availability of information.

Revenue can be forecasted using a top-down or bottom-up approach. Common top-down forecast objects include "growth relative to GDP growth" and "market growth and market share". Bottom-up drivers for revenue forecast include: volumes and average selling price, product-line or segment revenues, capacity-based measures, return or yield based measures.

LO: Explain approaches to forecasting a company's operating expenses and working capital.

Operating costs include cost of goods sold (COGS) and selling, general, and administrative expenses (SG&A). Disclosures about operating costs are often less detailed than revenue. Therefore, analysts are often forced to use more aggregated forecasted objects on the company level (rather than forecasting costs separately for different geographic regions, or business segments) or summary measures like EBITDA margins. However, forecasts for operating expenses should still be coherent with revenue forecasts.

Working capital forecasts are typically made using efficiency ratios as the forecast objects, which are combined with sales and cost forecasts to project accounts receivables,

inventories, accounts payable, and other current asset and liabilities.

LO: Explain approaches to forecasting a company's capital investments and capital structure.

Forecasts for capital expenditure may differentiate between maintenance and growth capital expenditures. Forecasts for maintenance capital expenditures are usually based on depreciation and amortization expenses. Forecasts for growth capital expenditures are linked to a company's strategy, expansion plans, and revenue growth.

Forecasts for a company's capital structure take into account the historical leverage ratios and capital structure, the company's financial strategy, and capital expenditure forecasts.

LO: Describe the use of scenario analysis in forecasting.

Rather than develop single point estimate forecasts, analysts make several forecast scenarios that vary based on different outcomes with respect to key risk factors. To make investment decisions, investors compare these scenarios to other analysts' forecasts for a company, as well as forecasts implied by current valuations.